

Sequence Analysis

Exercises

1. Consider the following table of sequential data taken over a 3 day period. Each day, the events are recorded in brackets, {}. If no events happened for that day, empty brackets are recorded. We're only interested in sequences of events that happened over a two day period (e.g. the rule $A \implies B$ would suggest that if A happens and then B happens the following day). For the rules listed below the table report the support, confidence, and pseudolift. How would these numbers change if we were interested in a three day time window?

UserID	Sequence
0001	{F,D} {B} {A,C}
0002	{B} {A,C} {F}
0003	{F} {D} {A,C}
0004	{B} {C} {}
0005	{D} {B} {A}
0006	{} {F} {B}
0007	{A} {} {F,C}
0008	{F} {B} {A}
0009	{B} {} {F,D}
0010	{F,D} {B} {A,C}

a) $F \implies A$

b) $B \implies A$

c) $D \implies C$

List of Key Terms

support

confidence

lift

pseudolift

sequence rule

association rule

transaction dataset

antecedent

consequent